Switch Statement

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The Java switch statements executes one statement from multiple conditions. It is like if-else-if ladder statement. The switch statement works with byte, short, int, long, enum types, String and Wrapper types like Byte, Short, Integer and Long.
In other words, **the switch statement "test the *equality* of a variable"** against multiple values.

Points to Remember

- There can be one or N number of case values for a switch expression.
- The case value must be of switch expression type only. The case value must be literal or constant. It doesn't allow variables.
- The case values must be unique. In case of duplicate values, it renders compile time error.
- The java switch expression must be of type byte, short, int long and string it also supports Wrapper types Byte, Short, Integer, Long.
- Each case statement can have a break statement which is optional. When control reaches to the break statement, it jumps the control after the switch block. If a break statement is not found, it executes the next case.
- The case value can have a default label which is optional.

Syntax:

```
switch(expression)
{
    case value1:
        //code to be executed
        break; //Optional
    case value2:
        //code to be executed
        break; //Optional
    case value3:
        //code to be executed
        break; //Optional
    …
    …
    default:  //optional
        Code to be executed if all cases are not matched;
}
```

```
package Chapter3;
import java.util.*;


/**
 * Write a description of class SwitchDemo here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class SwitchDemo
{
   public static void main(String args[]){
      Scanner sc = new Scanner(System.in);
      int monthNum;
      System.out.print("\nEnter month number(1-12): ");
      monthNum = sc.nextInt();
      switch(monthNum){
         case 1:{
```

```
            System.out.print("\nJanuary");
            break;
        }
        case 2:{
            System.out.print("\nFebruary");
            break;
        }
        case 3:{
            System.out.print("\nMarch");
            break;
        }
        case 4:{
            System.out.print("\nApril");
            break;
        }
        case 5:{
            System.out.print("\nMay");
            break;
        }
        case 6:{
            System.out.print("\nJune");
            break;
        }
        case 7:{
            System.out.print("\nJuly");
            break;
        }
        case 8:{
            System.out.print("\nAugust");
            break;
        }
        case 9:{
            System.out.print("\nSeptember");
            break;
        }
        case 10:{
            System.out.print("\nOctober");
            break;
        }
        case 11:{
            System.out.print("\nNovember");
            break;
        }
        case 12:{
            System.out.print("\nDecember");
            break;
        }
        default:{
            System.out.print("\nWrong month number!!!");
        }
    }
  }
}
```

If break statement is avoided then it will execute the next case statement. For example:

```
package Chapter3;
import java.util.*;
```

```
/**
 * Write a description of class CheckVowelOrConso here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class CheckVowelOrConso
{
   public static void main(String args[]){
      Scanner sc = new Scanner(System.in);
      char c;
      System.out.print("\fEnter any alphabet: ");
      c = sc.next().charAt(0);
      if((c >='A' && c <= 'Z') || (c >= 'a' && c <= 'z')){
         switch(c){
            case 'A': case 'E': case 'I': case 'O': case 'U':
            case 'a': case 'e': case 'i': case 'o': case 'u':
               System.out.print("\nEntered character is a vowel.");
               break;
            default:
               System.out.print("\nEntered character is a consonant.");
         }
      }
      else{
         System.out.print("\nEntered character is not an alphabet.");
      }
   }
}
```

If the user enters multiple character in the input only the first character is selected, which is starting from 0th position.

fall through cases: here the execution control will fall through if any of the case is true.

Basic Arithmetic Calculator

```
package Chapter3;
import java.util.*;


/**
 * Write a description of class BasicCalculator here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class BasicCalculator
{
   public static void main(String args[]){
      Scanner sc = new Scanner(System.in);
      int x, y, r = 0;
      String op;
      System.out.print("\f(+,-, *, /, %, ^, max, min) : ");
      op = sc.next();
      switch(op){
         case "+":{
            System.out.print("\nEnter first number: ");
            x = sc.nextInt();
            System.out.print("Enter second number: ");
            y = sc.nextInt();
            r = x + y;
            break;
         }
         case "-":{
            System.out.print("\nEnter first number: ");
```

```
            x = sc.nextInt();
            System.out.print("Enter second number: ");
            y = sc.nextInt();
            r = x - y;
            break;
        }
        case "*":{
            System.out.print("\nEnter first number: ");
            x = sc.nextInt();
            System.out.print("Enter second number: ");
            y = sc.nextInt();
            r = x * y;
            break;
        }
        case "/":{
            System.out.print("\nEnter first number: ");
            x = sc.nextInt();
            System.out.print("Enter second number: ");
            y = sc.nextInt();
            r = x / y;
            break;
        }
        case "%":{
            System.out.print("\nEnter first number: ");
            x = sc.nextInt();
            System.out.print("Enter second number: ");
            y = sc.nextInt();
            r = x % y;
            break;
        }
        case "max": case "Max": case "MAX":{
            System.out.print("\nEnter first number: ");
            x = sc.nextInt();
            System.out.print("Enter second number: ");
            y = sc.nextInt();
            r = Math.max(x, y);
            break;
        }
        case "min": case "Min": case "MIN":{
            System.out.print("\nEnter first number: ");
            x = sc.nextInt();
            System.out.print("Enter second number: ");
            y = sc.nextInt();
            r = Math.min(x, y);
            break;
        }

        case "^":{
            System.out.print("\nEnter first number: ");
            x = sc.nextInt();
            System.out.print("Enter second number: ");
            y = sc.nextInt();
            r = (int)Math.pow(x, y); //type-casting is done here from double to int
            break;
        }
        default:
            System.out.print("\nYou have selected wrong option from the menu.");
    }
    System.out.print("\nResult: " + r);
}
```

H.W

take float inputs to decimal operation, where it is required.

```
}
```